

Math 104 – Real Analysis

Comprehensive exercise solutions

Ross, *Elementary Analysis: The Theory of Calculus* (2nd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{8.6} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

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Chapter 1 Introduction

Ross, §§1–6: natural, rational, and real numbers; the completeness axiom

Chapter 2 Sequences

Ross, §§7–16: limits of sequences, limit theorems, monotone/Cauchy, subsequences, limsup/liminf, series

Chapter 3 Continuity

Ross, §§17–22: continuous functions, properties, uniform continuity, limits of functions

Chapter 4 Sequences and Series of Functions

Ross, §§23–27: power series, uniform convergence, term-by-term calculus

Chapter 5 Differentiation

Ross, §§28–31: derivative basics, Mean Value Theorem, L'Hospital, Taylor's Theorem

Chapter 6 Integration

Ross, §§32–36: Riemann integral, its properties, Fundamental Theorem of Calculus

Chapter 7 Capstone

Ross, §§37–38: exponents and logarithms; continuous nowhere-differentiable functions